

JARNASKLUBB, Sweden

WMO: 022840

Lat: 63.435N

Lon: 19.673E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-17.1	-22.5	0.5	-19.8	-19.8	0.6	-16.9	15.8	0.0	14.7	-0.6	3.8	350	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	22.4	18.5	20.9	17.4	19.6	16.2	19.5	21.8	18.1	20.1	17.0	18.9	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	21.2	17.2	12.3	19.5	16.0	11.4	18.2	55.6	21.9	51.2	20.2	47.7	18.9	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.3	12.3	10.5	DB	-22.2	24.3	4.2	1.6	-25.2	25.4	-27.7	26.4	-30.0	27.2	-33.0	28.4	(4)
(5)			WB	-22.6	19.6	3.8	1.3	-25.4	20.5	-27.6	21.3	-29.8	22.0	-32.5	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-4.5	-5.5	-2.4	2.0	7.3	12.4	16.1	15.0	10.9	5.2	1.0	-2.2	(6)
(7)		DBStd	8.32	5.49	6.05	4.31	2.75	3.19	2.56	2.38	2.47	2.80	3.41	4.19	5.17	(7)
(8)		HDD10.0	2435	449	434	383	241	96	7	0	1	23	151	272	379	(8)
(9)		HDD18.3	5002	707	667	641	491	341	179	79	107	224	407	522	638	(9)
(10)		CDD10.0	488	0	0	0	0	13	79	188	156	49	2	0	0	(10)
(11)		CDD18.3	13	0	0	0	0	0	1	9	4	0	0	0	0	(11)
(12)		CDH23.3	18	0	0	0	0	1	1	15	2	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.6	5.4	4.9	4.5	4.0	4.0	4.1	4.0	4.2	4.5	4.9	5.5	5.6	(14)
(15)	Precipitation	PrecAvg	591	40	31	29	34	39	49	69	69	59	59	62	45	(15)
(16)		PrecMax	903	73	60	46	84	73	78	176	146	147	148	150	80	(16)
(17)		PrecMin	430	9	8	10	6	5	26	11	8	20	14	16	15	(17)
(18)		PrecStd	101	17	14	11	21	16	17	41	41	33	34	34	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	6.5	8.9	12.3	18.7	21.7	25.2	23.2	18.1	12.9	9.6	6.5	(19)
(20)			MCWB	3.1	3.6	4.7	7.3	12.8	15.9	20.5	19.5	13.4	10.4	7.5	4.8	(20)
(21)		2%	DB	3.5	4.2	6.4	9.3	15.9	19.4	23.1	21.7	16.8	11.6	7.9	5.3	(21)
(22)			MCWB	2.3	2.1	3.1	5.4	11.0	14.1	19.0	18.5	13.7	9.8	7.1	4.3	(22)
(23)		5%	DB	2.7	2.7	4.3	7.5	14.0	17.9	21.7	20.4	15.9	10.9	7.1	4.5	(23)
(24)			MCWB	1.7	1.2	1.8	4.3	9.7	13.5	18.1	17.2	13.5	9.4	6.3	3.6	(24)
(25)		10%	DB	1.9	1.5	2.7	6.0	12.3	16.6	20.3	19.1	15.0	10.2	6.1	3.7	(25)
(26)			MCWB	1.1	0.6	1.0	3.5	8.6	12.9	17.2	16.1	13.1	8.5	5.3	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.7	5.1	7.5	13.6	16.8	21.2	20.9	15.7	11.6	8.0	5.5	(27)
(28)			MCDB	4.6	5.8	8.3	11.3	17.4	19.6	23.5	22.5	16.8	12.1	8.6	5.9	(28)
(29)		2%	WB	2.6	2.5	3.4	5.9	11.7	15.2	20.0	19.2	14.8	10.9	7.3	4.6	(29)
(30)			MCDB	3.3	4.0	5.9	8.6	15.2	18.0	22.4	21.1	15.9	11.2	7.9	5.1	(30)
(31)		5%	WB	1.9	1.5	2.1	4.8	10.3	14.2	18.8	17.8	14.1	9.9	6.4	3.7	(31)
(32)			MCDB	2.6	2.3	3.8	7.0	13.2	16.8	21.1	19.5	15.2	10.7	7.0	4.3	(32)
(33)		10%	WB	1.2	0.7	1.2	3.9	9.0	13.5	17.6	16.7	13.4	8.9	5.5	3.0	(33)
(34)			MCDB	1.8	1.4	2.4	5.5	11.7	15.8	19.7	18.4	14.5	9.9	6.1	3.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.9	6.1	5.6	6.0	5.9	5.5	5.6	5.3	4.5	3.7	4.6	(35)
(36)			MCDBR	4.3	5.4	7.2	8.3	8.4	7.8	7.2	6.7	6.1	4.6	3.3	3.9	(36)
(37)			MCWBR	3.9	4.2	4.8	5.2	4.7	4.5	4.2	4.3	4.3	3.9	3.0	3.6	(37)
(38)		5% WB	MCDBR	4.1	5.0	6.5	7.9	7.9	6.4	6.0	5.8	4.7	3.8	3.1	3.8	(38)
(39)			MCWBR	3.8	4.2	4.7	5.2	4.9	4.3	4.0	4.0	3.8	3.5	3.0	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.289	0.312	0.351	0.352	0.349	0.368	0.358	0.327	0.312	0.369	0.349	(40)	
(41)		taud	2.051	2.188	2.260	2.326	2.451	2.496	2.466	2.489	2.553	2.412	2.672	2.606	(41)	
(42)		Ebn at Noon	379	634	780	827	863	873	842	818	779	621	354	207	(42)	
(43)		Edh at Noon	29	61	85	100	96	94	95	85	66	50	27	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.15	0.74	2.23	3.76	5.00	5.67	5.21	3.95	2.38	0.96	0.24	0.07	(44)	
(45)		RadStd	0.02	0.14	0.25	0.35	0.54	0.39	0.51	0.34	0.26	0.17	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page